MMSE Pair Scheduling: Sharp Complexity, Certified Power Allocation, and Regime-Dependent Rate Gains

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***Abstract*—We study joint user pairing and power allocation in an uplink with linear minimum mean-square error (MMSE) reception and no successive interference cancellation. Each user is served once, two users share each resource, and positive bounded powers satisfy an aggregate budget. We prove scheduling hardness with three receive dimensions and a fixed rational unit-channel alphabet, derive the sharp whole-quadrant concavity threshold, and characterize rectangular power pricing by at most seventeen candidates. For network optimization, we combine a finite-box/integral certificate with an exact resource-labelled dynamic program and successive convex approximation of the powers. On 32 stored ray-traced networks at $M = 8$, the certificate attains a verified gap of at most 0.01 nat in 27 cases, with a 97.82% median reduction in the fixed-anchor gap. A receive-dimension audit explains why earlier rate gains were small: at $M = 8$, MMSE reception nearly nulls the two-user interference and the fixed-power pairing gain is zero in all 32 cases. Headroom appears only at small M and large K. On 53 standardized 3GPP TR 38.901 CDL-A/B/C cases at $K = 16$ and $M = 2$, the exact search improves the strongest common-model baseline by a median 1.31%, or 0.129 bit/s/Hz, with 53 wins in 53 cases and a 95% interval $[1.19, 1.42]\%$. At $K = 8$ it matches SLSQP with a negligible absolute gain while running about 211 times faster. The CDL rate gains are feasible and verified but not globally certified; the certified result remains the $M = 8$ bound.**

***Index Terms*—Uplink scheduling, MMSE reception, power allocation, global optimization, computational complexity, receive-dimension regimes.**

I. Introduction

UPLINK pair scheduling lets two users share a resource while a linear minimum mean-square error (MMSE) receiver separates their signals without successive interference cancellation (SIC). The two-user rate depends on the channel alignment and on both powers. A power update changes the receiver, and a pairing update changes which interference the receiver must suppress. The resulting problem combines an integral assignment with continuous power allocation under a shared budget.

The two-user no-SIC rate is established in [1]. Fan et al. [2] optimize pairing at prescribed powers, and Ruder et al. [3] also examine imperfect scheduling information. Matching-based fractional programming [4] combines scheduling with continuous updates, while weighted-MMSE precoding [5] accounts for multi-antenna cell-free processing and error covariance. Recent no-SIC and dual-polarized designs [6] use a different receiver and finite-blocklength model. These studies establish the architecture and the rate model. They do not provide a bound on how far a computed integral schedule and its continuous powers are from the optimum.

A feasible schedule gives an achieved rate. Assessing its distance from the optimum also requires an upper bound. Global optimization frameworks, including mixed-monotone optimization [7], provide such bounds at additional computational cost. We use the exact structure of the two-user MMSE rate to make the bound practical. Two difficulties arise: a power grid can omit better allocations, and a continuous configuration relaxation can mix assignments or powers that one integral schedule cannot realize.

The first contribution is a structural analysis. We prove that fixed-power scheduling is NP-complete with three receive dimensions and rational unit channels from a fixed finite alphabet. A continuous-power extension retains a constant additive separation. We derive the sharp whole-quadrant concavity condition for the pair log-rate and characterize rectangular power pricing by at most seventeen candidates. The bound uses a sufficient finite-box concavity test, verified rectangular partitions and exact resource-labelled dynamic programming.

The second contribution is an exact joint search built on the same structure. At fixed powers, the integral assignment is solved exactly by subset dynamic programming. At fixed assignment, the continuous powers are refined by successive convex approximation with a recomputed MMSE receiver. Alternating these two exact subproblems from several starts gives a feasible schedule at a small fraction of the time required by a full SLSQP assignment search.

The third contribution is a regime analysis. The original 32-case benchmark uses $M = 8$ receive dimensions and two users per resource. At that dimension, MMSE reception nearly nulls the two-user interference, and the fixed-power pairing gain is zero in all 32 cases. The certified headroom over the grid is also negligible in the O1 scene. This explains why prior rate comparisons were flat: the benchmark had little pairing headroom to exploit. Truncating the same stored channels to $M = 2$ and $M = 4$ restores pairing headroom. The effect is monotone: the median pairing gain is 8.87% at $M = 2$, 2.61% at $M = 4$, and 0.00% at $M = 8$.

The fourth contribution is an evaluation of the exact search on a standardized CDL benchmark. On 53 3GPP TR 38.901 CDL-A/B/C cases at $K = 16$ and $M = 2$, the proposed method improves the strongest common-model baseline by a median 1.31%, or 0.129 bit/s/Hz, with 53 wins in 53 cases and a 95% interval $[1.19, 1.42]\%$. At $K = 8$ it matches SLSQP with a negligible absolute gain while running about 211 times faster by the ratio of median times. The gain is modest in relative terms because the sum rate grows with K, but it is material in absolute terms and statistically resolved.

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We also report the boundary of the certificate. The original $M = 8$ result remains 27 of 32 certificates at 0.01 nat with a 97.82% median fixed-anchor gap reduction. At $M = 2$, only 10 of 32 cases retain tight certificates; high-SNR and large-K cases leave gaps of 0.25–30 nat at the tested split cap. The rate gains in that regime are therefore feasible and verified but not globally certified. We state this boundary explicitly. The paper does not claim certified rate superiority at $M = 2$, and it does not claim broad superiority at $M = 8$.

Section II reviews related work. Section III gives the model, Section IV the structural analysis, and Section V the certified network optimization. Section VI reports the receive-dimension audit. Section VII presents the exact search. Section VIII gives the CDL benchmark and results, and Section IX discusses limitations.

II. Related Work

A. Pair scheduling and computational complexity

Ruder et al. [8] distinguish homogeneous-resource pairing from frequency-selective grouping. We consider the additional coupling introduced by continuous powers and an aggregate budget, and seek bounds for an integral full-service schedule.

Mostafa et al. [9, Th. 4] encode three-dimensional matching (3DM) through channel orthogonality for decorrelator-SIC offloading, using dimension $L = |X \times M| + 2$ that grows with the instance. Their model includes admission, delay and energy constraints. We establish hardness for exact-once no-SIC sum-rate maximization using a fixed-dimensional rational channel alphabet. The receiver, objective and service constraints distinguish the two results.

Related hardness results cover LTE-A scheduling [10], SIC allocation [11], [12], and fixed-antenna interference alignment and transceiver design [13], [14]. Our result combines a fixed receive dimension, a fixed rational unit-channel alphabet and the stated full-service constraints.

B. Power optimization and bounds

Shen and Yu's quadratic transform [15] gives an exact auxiliary reformulation with stationary-point convergence under its assumptions. Their matching-based scheduling [4] and Wang et al.'s weighted-MMSE precoding [5] motivate our common-model comparator. Shi et al. [16] establish the rate–MMSE relation and globally solved fixed-filter power steps. Our pairwise pricing retains receiver recomputation.

Ul Hassan et al. [17] derive finite candidates for two-link power sharing with fixed interference gains and an exhausted sum budget. Sharma and Teneketzis [18] obtain decentralized guarantees for concave utilities, including negative MMSE. We study the log-rate with recomputed MMSE reception and specialize mixed-monotone optimization [7] to this utility. Barnhart et al.'s branch-and-price framework [19] requires complete or bounded pricing. Zhao and Pottie's divisible-spectrum convexification [20] permits sharing across allocations, whereas our schedule is integral.

C. Recent resource-allocation designs

Ou et al. [6] transmit two streams per dual-polarized user in a no-SIC uplink, accounting for estimation errors and a finite-blocklength dispersion penalty. Their Algorithm 1 fixes combiners before the power loop. Here, we recompute MMSE filters as powers change and optimize integral pair/resource assignments under an aggregate budget.

Chen et al. [21] jointly optimize downlink grouping, subcarriers and hybrid beamforming. Their beam-squint and radio-frequency constraints concern a different transmission architecture.

Fu et al. [22] alternate downlink power and LP-RZF regularization using statistical CSI. Thu et al. [23] combine satellite/cell-free load balancing with differential-evolution power search for energy efficiency. We adapt their power surrogate and search components to our sum-rate objective and recomputed receiver; the source architectures and guarantees remain specific to those studies. The Thu adaptation follows arXiv:2607.21860v1. Supplementary Section S7 gives the adaptations in detail.

Our earlier work addresses user clustering and association [24] and learned channel estimation [25]. The present study focuses on pair scheduling and power allocation at a common receiver with specified channels. Extending its bounds to multi-AP processing, SIC or uncertain channel estimates requires corresponding changes to the model.

III. System Model and Problem Formulation

A. Uplink transmission and information assumptions

We consider $K = 2S$ single-antenna users and a common M-dimensional receiver over S orthogonal resources. Exactly two users share each resource, and every user is served once. This full-service constraint includes users with unfavorable channels and separates pairing from admission control.

Let $\mathcal{K} = \{1, \dots, K\}$, $\mathcal{S} = \{1, \dots, S\}$, and $\mathcal{E} = \{\{u, v\} : u < v,\ u, v \in \mathcal{K}\}$. The channel of user i on resource s is $\mathbf{h}_{is} \in \mathbb{C}^M$. After whitening by the known positive-definite noise covariance, the received signal for pair $e = \{u, v\}$ is

$$\mathbf{y}_s = \sqrt{p_u}\mathbf{h}_{us}a_u + \sqrt{p_v}\mathbf{h}_{vs}a_v + \mathbf{n}_s, \qquad \mathbb{E}[\mathbf{n}_s\mathbf{n}_s^H] = \mathbf{I}_M. \quad (1)$$

A receive direction that maximizes the signal-to-interference-plus-noise ratio (SINR) of user u is proportional to $(\mathbf{I}_M + p_v\mathbf{h}_{vs}\mathbf{h}_{vs}^H)^{-1}\mathbf{h}_{us}$. The linear MMSE filter has the same direction, and its scale leaves the SINR unchanged. We retain this dependence on the other user's power.

The symbols $a_i \sim \mathcal{CN}(0, 1)$ are independent across users and independent of $\mathbf{n}_s \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$. We use Gaussian-input rates with residual interference treated as noise. Powers are in watts. For scalar noise, whitening divides each physical channel by the noise standard deviation, retaining the path loss already included in the channel.

The scheduler and receiver know the same channel realization. We use *nominal* to indicate that optimization and certification refer to this realization at its stored precision. Imperfect channel state information (CSI) would require an

additional uncertainty model. The allocation represents a central controller serving a fixed cohort in one interval under a prescribed aggregate radiated-power allowance.

Users satisfy $0 < p_{\min} \le p_i \le p_{\max}$ and a shared budget $\sum_i p_i \le P$. The controller imposes P across the scheduled users and resources, in addition to each user's bounds. The positive lower bound requires every assigned user to transmit. Since every pair is eligible, $Kp_{\min} \le P$ guarantees a feasible power allocation for any schedule. Resources have equal bandwidth and are mutually orthogonal. We maximize unweighted sum rate under these constraints, with no SIC or user-rate requirement.

Fig. 1 shows the uplink architecture and pair-rate calculation. The evaluated receiver has $M = 8$, while the theory permits general M. Pair identities and resource labels are decision variables.

B. Recomputed linear MMSE reception

For each selected pair, the receiver recomputes the linear MMSE filter at the selected powers and treats the other user's signal as noise. We use Haustein's two-user rate [1] to analyze curvature and joint allocation. Define

$$\begin{aligned} g_{is} &= \|\mathbf{h}_{is}\|^2 \ge 0, \\ r_{uvs} &= \frac{|\mathbf{h}_{us}^H \mathbf{h}_{vs}|^2}{g_{us} g_{vs}} \in [0,1] \quad (g_{us} g_{vs} > 0). \end{aligned} \tag{2}$$

Here r_{uvs} is the squared normalized channel correlation: zero denotes orthogonal channels and one denotes collinear nonzero channels. If either gain is zero, we set $r_{uvs} = 0$; the corresponding zero received power makes this convention consistent with the MMSE rates. The zero-channel user remains assigned and subject to the same power lower bound. With $x = g_{us}p_u$ and $y = g_{vs}p_v$, the matrix inversion lemma gives the SINRs

$$\gamma_u = x\frac{1+(1-r_{uvs})y}{1+y}, \qquad \gamma_v = y\frac{1+(1-r_{uvs})x}{1+x}. \tag{3}$$

For generic normalized powers $x, y \ge 0$ and squared correlation $r \in [0,1]$, define the pair sum rate in nats per channel use:

$$\begin{aligned} q_r(x,y) &= 2\ln d - \ln(1+x) - \ln(1+y), \\ d &= 1 + x + y + (1-r)xy. \end{aligned} \tag{4}$$

Since $1+\gamma_u = d/(1+y)$ and $1+\gamma_v = d/(1+x)$, summing their logarithms gives (4). This scalar form retains the receiver coupling and avoids repeated matrix inversion during objective evaluation.

We write $q_{uvs}(p_u, p_v) = q_{r_{uvs}}(g_{us}p_u, g_{vs}p_v)$. Formula (4) includes receiver recomputation; fixing receiver coefficients while varying powers gives a different objective.

Let $e_s \in \mathcal{E}$ be the pair selected on resource s. The aggregate sum of per-resource spectral efficiencies (SEs) is

$$Q = \frac{1}{\ln 2}\sum_{s=1}^{S} q_{e_s,s}. \tag{5}$$

Thus Q sums bit/channel-use values across resources, and gross occupied-band SE is Q/S bit/s/Hz. For resource bandwidth B_{r} and data fraction η, throughput is $T = \eta B_{\mathrm{r}} Q$ bit/s and net occupied-band SE is $\eta Q/S$. Throughput evaluation requires specified bandwidth and overhead; energy efficiency additionally requires amplifier, circuit and processing-power models.

C. Joint continuous problem

Let $z_{uvs} \in \{0,1\}$ indicate the pair placed on resource s. The joint problem is

$$\begin{aligned} \mathcal{P}_{\mathrm{C}}: \quad \max_{z,p} \quad & \sum_{s\in\mathcal{S}}\sum_{u<v} z_{uvs} q_{uvs}(p_u,p_v) \\ \text{s.t.} \quad & \sum_s \Big(\sum_{v<i} z_{vis} + \sum_{v>i} z_{ivs}\Big) = 1 \ (i \in \mathcal{K}), \\ & \sum_{u<v} z_{uvs} = 1 \quad (s \in \mathcal{S}), \\ & p_{\min} \le p_i \le p_{\max} \quad (i \in \mathcal{K}), \\ & \sum_i p_i \le P \end{aligned} \tag{6}$$

The first two equalities enforce service once per user and one pair per resource. Each power is indexed by its user, whose rate also depends on the selected partner and resource. Problem (6) therefore couples discrete scheduling with continuous powers; the scheduling decisions remain even when every pair utility is concave.

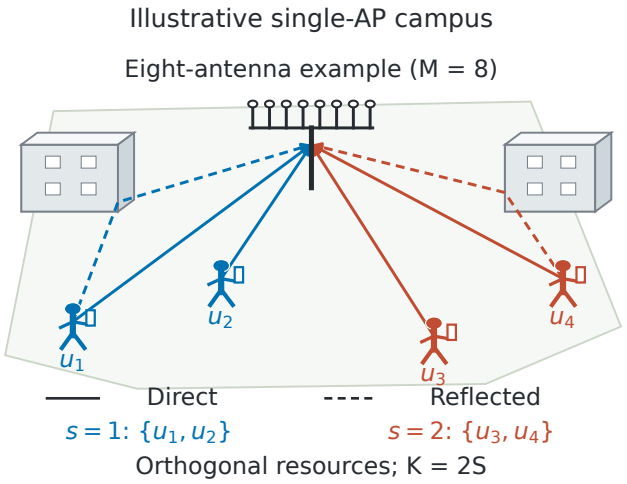
Viterbo and Hottinen [26] distinguish fixed-power disjoint pairing from an expanded cyclic scheduling family. With fixed powers and resource-invariant channels, our weights q_{uvs} are independent of s, and maximum-weight perfect matching solves the scheduling step. Edmonds' general-graph framework [27] enforces disjoint pairs; unrestricted bipartite assignment need not do so. Resource-dependent channels require explicit resource labels.

D. Finite-grid integer and linear programs

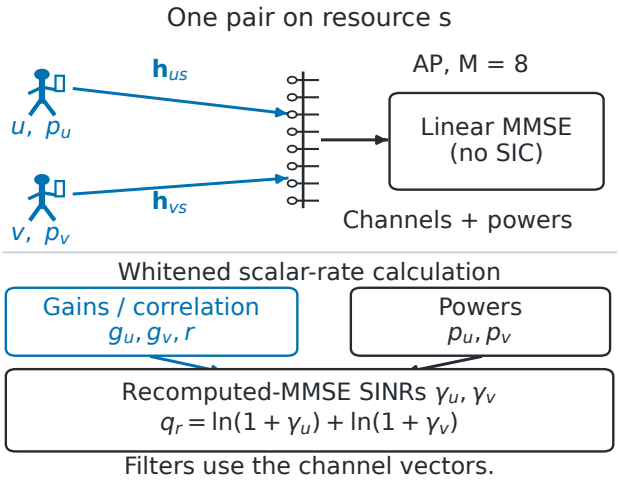
Let $\mathcal{G} \subset [p_{\min}, p_{\max}]$ be a prescribed finite power grid. A configuration $c = (u, v, s, p_u, p_v)$ includes its resource, pair and two powers. Let $\mathcal{C}_{\mathcal{G}}$ contain all such configurations, $a_{ic} = 1$ when user i belongs to c, $b_{sc} = 1$ when c uses resource s, $d_c = p_u + p_v$, and $q_c = q_{uvs}(p_u, p_v)$. The grid integer program (IP) is

$$\begin{aligned} \mathcal{P}_{\mathrm{IP}}(\mathcal{G}): \quad \max_{z} \quad & \sum_c q_c z_c \\ \text{s.t.} \quad & \sum_c a_{ic} z_c = 1, \quad i \in \mathcal{K}, \\ & \sum_c b_{sc} z_c = 1, \quad s \in \mathcal{S}, \\ & \sum_c d_c z_c \le P, \qquad z_c \in \{0,1\} \end{aligned} \tag{7}$$

The grid linear program (LP), $\mathcal{P}_{\mathrm{LP}}(\mathcal{G})$, replaces integrality with $z_c \ge 0$. The incidence equalities imply $z_c \le 1$. Its dual



(a) Full-service uplink architecture.



(b) Recomputed-MMSE pair calculation.

Fig. 1. Uplink architecture and MMSE pair-rate calculation. Two users share each orthogonal resource at a common receiver without SIC. The evaluated access point has eight antennas; the geometry and propagation paths are schematic. Gains, correlation and powers determine the pair rate, while filter construction uses the full channels.

uses unrestricted user prices α_i, unrestricted resource prices β_s, and a nonnegative power price λ:

$$\begin{aligned} \min_{\alpha,\beta,\lambda} \quad & \sum_i \alpha_i + \sum_s \beta_s + \lambda P \\ \text{s.t.} \quad & \alpha_u + \alpha_v + \beta_s + \lambda(p_u + p_v) \geq q_{uvs}(p_u, p_v),. \\ & c \in \mathcal{C}_\mathcal{G}, \\ & \lambda \geq 0 \end{aligned} \tag{8}$$

The grid LP upper-bounds the grid IP, but may lie below the continuous optimum in (6) because powers outside $\mathcal{G}$ can yield better allocations. A feasible grid IP schedule also satisfies (6) and supplies a lower bound after rate verification. Bounding the continuous optimum requires coverage of the full power domain.

Replacing grid weights by nonnegative Borel measures on each pair's power rectangle gives the continuous configuration LP: incidence masses equal one, the total power moment is at most P, and the objective integrates q_{uvs}. Its formal primal is (D9) in Supplementary Section S4; its dual imposes (8) at every power. Write V_{CLP} for its value and V_{C} for (6). Dirac masses embed integral schedules, so $V_{\mathrm{C}} \leq V_{\mathrm{CLP}}$; assignment and power mixtures can make the inequality strict.

IV. Structural Analysis

A. Hardness with a fixed receiver dimension

Decision problem IDEAL-MMSE-PAIR. Fix the rational unit-vector alphabet $\mathcal{A} \subset \mathbb{Q}^3$ constructed in Supplementary Section S1. An instance gives $K = 2S$ and an explicit $K \times S$ table of alphabet indices, with unit powers and unit white-noise variance. Does a full-service pair/resource schedule attain at least $K \ln 2$? The threshold is specified by K, not a numerical logarithm. Malformed encodings are rejected.

Theorem 1. IDEAL-MMSE-PAIR is NP-complete. Thus fixed-power sum-rate maximization is NP-hard with $M = 3$ and the fixed rational unit alphabet $\mathcal{A}$. For fixed rational $0 < p_{\min} < \bar{p} < p_{\max}$, the continuous-power family with $P = K\bar{p}$ is also NP-hard through ideal-threshold attainment. Both constructions have a positive constant additive separation in natural-log sum-rate units.

Proof sketch. Use perfect 3DM with each element in at most three triples [28, Sec. 2, (2.1)]. Users are $X \cup Y$ and resources are Z; each resource's admissible graph has at most three edges and is a forest. Supplementary Section S1 realizes exactly these edges as zero inner products, including an outsider role. A unit-power pair reaches $2 \ln 2$ exactly when orthogonal. A schedule witness has $O(K \log K)$ bits; the rational product $\prod_i (1 + \gamma_i) \geq 2^K$ is checked with $O(K)$-bit numerator and denominator because the alphabet is fixed. For continuous powers, Jensen bounds the ideal sum by $K \ln(1 + \bar{p})$; a forbidden pair has a fixed positive loss. Supplementary Section S4 gives the reduction size and explicit gaps (D1)–(D3). □

In the continuous family, the fixed power constants and ideal threshold $K \ln(1 + \bar{p})$ are represented by their defining rational data and K. Theorem 1 combines the restrictions stated above; Section II compares them with the channel-orthogonality construction in [9].

The separation is additive. A constant relative approximation gap, NP-membership for unrestricted continuous thresholds and a receive-dimension tractability boundary remain outside Theorem 1. The construction permits channels to vary independently across resources and assumes no common-ray geometry. AIM [29, Th. 3.16, p. 1639] provides the real tree-complement representation; Supplementary Section S1 constructs the rational unit alphabet for the bounded forest family.

B. Sharp pairwise curvature

Theorem 2. The function q_r in (4) is jointly concave over $\mathbb{R}^2_+$ if and only if

$$r \leq 1 - \frac{1}{\sqrt{2}}. \tag{9}$$

Within this range $\nabla^2 q_r(x,y) \prec 0$ at every finite $(x,y) \in \mathbb{R}^2_+$.

Proof sketch. Let $a = 1 - r$. At $x = 0$, $q_{xx} \to 1 - 2a^2$ as $y \to \infty$, proving necessity by continuity to an interior point. For $a \geq 1/\sqrt{2}$, both diagonals are nonpositive and the determinant numerator has nonnegative coefficients with positive constant term. Thus the Hessian is negative definite. Supplementary Section S4, (D4)–(D5), gives the Hessian and numerator; Section S2 lists every coefficient. □

This squared-correlation threshold applies to the whole nonnegative quadrant. A finite rectangle can remain concave at higher correlations, so local curvature also depends on the operating powers.

C. Complete rectangular power pricing

To extend a configuration dual from grid points to continuous powers, we must bound the rate minus its power cost over every pair rectangle. We call this maximization power pricing:

$$\begin{aligned} H = \max_{(x,y)\in\mathcal{B}} \{q_r(x,y) - \lambda_1 x - \lambda_2 y\}, \\ \mathcal{B} = [L_x, U_x] \times [L_y, U_y]. \end{aligned} \tag{10}$$

For positive gains and a network price λ in natural-log rate units, $\lambda_1 = \lambda/g_{us}$ and $\lambda_2 = \lambda/g_{vs}$. The following result also allows signed prices, although the network power multiplier is nonnegative.

Theorem 3. For $0 < r \leq 1$, arbitrary real λ_1, λ_2, and a compact rectangle in $\mathbb{R}^2_+$ with $L_x < U_x$ and $L_y < U_y$, the maximum in (10) occurs among at most seventeen points: four corners, at most two stationary points on each of four edges, and at most five interior stationary points.

Proof sketch. Put $a = 1 - r$, $A = 1 + x$, $k = 1 + ax$, and

$$C = A(1 - \lambda_1 A), \qquad B = 2aA - k(1 + \lambda_1 A).$$

Multiplying $q_x = \lambda_1$ by the positive Ad gives $C + By = 0$. If $C = 0$, then $B = -2r \neq 0$; hence a stationary point has $B \neq 0$ and $y = -C/B$. Substituting into $q_y = \lambda_2$ yields

$$\begin{aligned} \Pi(x) = [2k - (1 + \lambda_2)A]B^2 \\ - [k - \lambda_2(A + k)]CB - \lambda_2 k C^2. \end{aligned}$$

This polynomial has degree at most five and is nonzero because $\Pi(-1) = 2r^3 > 0$. Retain only roots with $B \neq 0$, $(x, -C/B)$ strictly inside the rectangle, and both original equations satisfied. Each edge has a nonzero polynomial of degree at most two; corners cover its endpoints. Thus four corners, eight edge candidates and five interior candidates suffice. Supplementary Section S4 supplies the edge equation and full admissibility argument. □

For $r = 0$ the objective is separable. For $j = x, y$, set $(\lambda_x, \lambda_y) = (\lambda_1, \lambda_2)$ and use $\mathrm{clip}(1/\lambda_j - 1, [L_j, U_j])$ if $\lambda_j > 0$, or U_j otherwise, where $\mathrm{clip}(t, [L, U]) = \min(U, \max(L, t))$. Collapsed rectangles reduce to an edge or point. A zero-gain user's physical contribution is $-\lambda p_i$, maximized at minimum power for $\lambda \geq 0$; the other coordinate is a scalar logarithm. Additional coupled pair constraints require their own boundary analysis.

Theorem 3 gives a finite candidate characterization. Certification requires isolation of all relevant real roots and directed logarithmic enclosures, with computational cost depending on precision. An unresolved enclosure leaves the pricing calculation uncertified. The recomputed-MMSE objective distinguishes this result from earlier elimination methods [17].

V. Certified Network Optimization

A. Repairing a numerical configuration dual

The grid master imposes pricing constraints only at its configurations. We raise each resource price to cover every continuous configuration, obtaining a valid upper bound from arbitrary candidate user prices and a nonnegative power price.

Proposition 1. Let α_i be arbitrary real values and $\lambda \geq 0$. Suppose H_{uvs} is a valid upper bound on

$$\max_{p_u, p_v \in [p_{\min}, p_{\max}]} \{q_{uvs}(p_u, p_v) - \lambda(p_u + p_v)\}. \tag{11}$$

Set $\beta_s = \max_{u<v}(H_{uvs} - \alpha_u - \alpha_v)$. Then

$$\begin{aligned} U_{\mathrm{dual}} = \sum_i \alpha_i + \sum_s \beta_s + \lambda P, \\ V_{\mathrm{C}} \leq V_{\mathrm{CLP}} \leq U_{\mathrm{dual}}. \end{aligned} \tag{12}$$

Proof. The choice of β_s makes every continuous configuration inequality in (8) valid. Multiply those inequalities by any feasible configuration mass and sum. User and resource equalities produce the first two sums in (12), while nonnegativity of λ bounds the power contribution by λP. For an integral schedule this is directly

$$\begin{aligned} \sum_s q_{e_s,s} &\leq \sum_s (\alpha_{u_s} + \alpha_{v_s} + \beta_s) + \lambda \sum_i p_i \\ &\leq \sum_i \alpha_i + \sum_s \beta_s + \lambda P. \end{aligned}$$

Each user price is counted once precisely because service is exact-once. This is weak duality and does not require that the candidate prices solve the numerical master exactly. □

Every pair/resource rectangle must be covered, including pairs absent from the numerical master. Let L be a verified lower enclosure of a feasible integral schedule's objective. Then $L \leq V_{\mathrm{C}} \leq U_{\mathrm{dual}}$ bounds its loss by $U_{\mathrm{dual}} - L$. We retain the smallest complete upper bound U among the fallback U_0, spatial and repaired-dual bounds, and the integral bound introduced below.

The continuous configuration relaxation can retain an integrality gap after dual repair. Supplementary Section S3 gives assignment and fixed-pair power examples. In the latter, $g_u = g_v = r = 1$, $p_i \in [1, 2]$ and $P = 11/5$. The deterministic optimum is $\ln(128/55)$, but masses $4/5$ at $(1, 1)$ and $1/5$ at $(1, 2)$ have the same average power and strictly larger objective. This power-mixing gap persists even after the assignment is fixed.

B. Complete small-network reference

There are $K!/2^{K/2}$ assignments of unordered user pairs to labelled resources. We enumerate this family for small networks. For each assignment, the remaining domain is a box in K physical power variables intersected with $\sum_i p_i \le P$. Subdivision of the power domain resolves the power-mixing gap as well as the assignment gap.

For a normalized pair box $[\ell_x, u_x] \times [\ell_y, u_y]$, let $a = 1 - r$. The mixed-monotone bound is

$$U_{\mathrm{MM}} = \ln\left(1 + u_x \frac{1 + a\ell_y}{1 + \ell_y}\right) + \ln\left(1 + u_y \frac{1 + a\ell_x}{1 + \ell_x}\right). \quad (13)$$

Each term increases in its own power and decreases in the interfering normalized power, proving the bound directly from (3). A budget-aware alternative decomposes $q = f + g_x + g_y$, where $f = 2\ln d$, $g_x = -\ln(1 + x)$ and $g_y = -\ln(1 + y)$. The Hessian of f is

$$\nabla^2 f = -\frac{2}{d^2}\begin{bmatrix}(1 + ay)^2 & r \\ r & (1 + ax)^2\end{bmatrix}. \quad (14)$$

The matrix inside (14) is positive semidefinite because $(1 + ax)(1 + ay) \ge 1 \ge r$. Hence f is concave and its tangent majorizes it. Both g_x, g_y are convex and lie below their endpoint chords. The tangent plus both chords is an affine upper bound; degenerate intervals use their point value.

Summing these affine bounds over the assignment and maximizing over the physical power box and total budget gives a continuous knapsack problem: start at the lower powers, allocate remaining budget to positive affine slopes in descending order, and stop at each upper bound. This upper bound retains the total-power constraint. Its minimum with the sum of the pair bounds in (13) is also valid. We evaluate the tangents, secants and knapsack arithmetic with outward bounds.

When the exact rational condition $(1 - r)^2 \ge 1/2$ holds, Theorem 2 supplies a supporting tangent for the entire pair rate. The curvature-aware reference uses this tighter affine bound in place of the tangent-secant construction, with the same budget maximization and mixed-monotone intersection. The remaining pairs use the generic bound.

The reference covers every assignment, selects a live box with largest bound and bisects a longest physical-power interval. It discards infeasible or dominated boxes and evaluates a budget-feasible center in each new feasible child. The maximum retained upper bound remains valid at interruption and may be intersected with U_{dual}. Supplementary Section S5 gives the complete procedure and finite work limits.

Proposition 2. In exact arithmetic, with finitely many assignments and the stated compact power domain, best-bound selection with longest-edge bisection and the stated bounds terminates at any tolerance $\varepsilon > 0$ if computation is unrestricted, every feasible singleton initial box is evaluated, and every newly created feasible child box is evaluated at a feasible point, with each evaluation updating the incumbent.

Proof. Evaluating a feasible singleton initial box makes its exact upper no larger than L. The remaining initial boxes form finitely many exceptions, each discarded or bisected when selected. For evaluated feasible boxes, the upper minus the point value vanishes uniformly with diameter on the compact domain. Longest-edge bisection has only finitely many nodes above any positive diameter threshold, and every feasible descendant updates L on creation. Thus only finitely many selected boxes can exceed $L + \varepsilon$; best-bound selection reaches $\max U_{\mathrm{live}} \le L + \varepsilon$. Supplementary Section S4 gives the full argument. □

The singleton-root initialization is a premise of the ideal procedure in Proposition 2. The archived implementation was measured with nondegenerate power intervals $[0.05, 0.2]$ W. Assignment enumeration, work limits and finite precision constrain that reference; its rectangular pricing is separate from Theorem 3's algebraic oracle.

For initialization, pair users $(2s - 1, 2s)$ on resource s and set every power to $p_0 = \min(p_{\max}, P/K)$. This is feasible when $Kp_{\min} \le P$. The common fallback is

$$U_0 = \sum_{i=1}^{K} \max_{s \in \mathcal{S}} \ln(1 + g_{is} p_{\max}), \qquad g_{is} = \|\mathbf{h}_{is}\|^2.$$

This ceiling relaxes interference, the aggregate budget and resource-incidence constraints, providing a valid initial upper bound.

A separate verifier reconstructs gains and correlations from stored binary64 channels, checks schedule and power constraints, and encloses logarithms using integer arithmetic with explicit remainders. Partition trees and dual weights establish coverage, with the interference-free ceiling covering unvisited assignments. The bounds apply to the stored nominal channels; Supplementary Sections S5 and S10 describe verification and reproduction.

C. Finite-box curvature and integral scheduling

We strengthen the network bound by testing curvature on the actual power boxes and retaining integral assignments at each power price. Complete rectangular pricing covers every pair/resource combination.

Proposition 3 (Concavity on a finite power box). Let $\mathcal{B} = [\ell_x, u_x] \times [\ell_y, u_y] \subset \mathbb{R}^2_+$ be finite, $0 \le r \le 1$, $a = 1 - r$ and $d(x, y) = 1 + x + y + axy$. Define

$$u_0 = \frac{(1 + \ell_x)(1 + au_y)}{d(\ell_x, u_y)}, \quad v_0 = \frac{(1 + \ell_y)(1 + au_x)}{d(u_x, \ell_y)}, \quad (15)$$
$$C_\ell = (1 + \ell_x)(1 + \ell_y), \quad C_u = (1 + u_x)(1 + u_y),$$

$$W_1 = \frac{rC_u}{d(\ell_x, \ell_y)^2}, \qquad W_2 = \frac{rC_u^2}{C_\ell d(u_x, u_y)^2}, \quad (16)$$
$$w_0 = \min\{W_1, W_2\}.$$

Then q_r is concave on $\mathcal{B}$ if

$$A_0 = 2u_0^2 - 1 \ge 0, \qquad B_0 = 2v_0^2 - 1 \ge 0, \quad (17)$$
$$A_0 B_0 \ge 4w_0^2.$$

Diagonal Hessian scaling proves sufficiency: u_0, v_0 are exact minima of the diagonal factors, and $2w_0$ bounds the mixed derivative's magnitude. The expression for W_2 has a positive denominator even at $r = 1$. The whole-quadrant condition $(1 - r)^2 \ge 1/2$ remains an independent alternative.

A qualifying box admits a supporting tangent at any point in it. We subtract the physical cost $\lambda(p_u + p_v)$ and maximize the affine upper over the box. Generic tangent–secant and mixed-monotone bounds cover the remaining boxes. Taking the largest leaf upper over a complete partition gives H_{uvs} in (11). Zero-gain powers retain their linear cost, and collapsed intervals use endpoint values. Supplementary Section S11 proves these cases and the coverage requirements.

Proposition 4 (Integral scheduling bound). Let $\mathcal{Z}_I$ contain all full-service integral schedules, $K = 2S$, and suppose every H_{uvs} satisfies (11) for one common $\lambda \geq 0$. Then

$$V_{\mathrm{C}} \leq U_I := \lambda P + \max_{z \in \mathcal{Z}_I} \sum_{s,u<v} H_{uvs} z_{uvs}. \tag{18}$$

Each feasible schedule counts every physical power once, so subtracting the price and using the budget proves the bound. Ordering the resources gives

$$D(s, A) = \max_{\substack{u<v \\ u,v \in A}} \{H_{uvs} + D(s+1, A \setminus \{u, v\})\}. \tag{19}$$

Here $s = 1 + (K - |A|)/2$, $D(S+1, \varnothing) = 0$, and infeasible states equal $-\infty$. The root uses all users. Every pair in A is considered for the current labelled resource; fixing its smallest user would omit schedules. The recurrence uses $O(K^2 2^K)$ arithmetic operations and $O(2^K)$ memory.

With validated rational uppers and integer scale $\Gamma = 2^{40}$, we use

$$\begin{aligned} \widehat{H}_{uvs} &= \lceil \Gamma H_{uvs} \rceil, \\ U_{\mathrm{cert}} &= \lambda P + \frac{\widehat{D}}{\Gamma} \geq U_I, \end{aligned} \tag{20}$$

where $\widehat{D}$ is the integer-weight root value and λP remains exact. Signed weight ceilings add less than S/Γ relative to the same unrounded table. Rational sign tests and outward logarithmic enclosures establish the pair uppers before integer rounding.

For the same λ, H, every repair in Proposition 1 satisfies

$$U_I \leq U_{\mathrm{dual}}. \tag{21}$$

The ordering requires a common price and pair-bound table; it need not hold across different numerical proposals or rounding slack. The bound concerns the integral optimum, and optimizing λ can retain the budget gap in Supplementary Section S3. The network implementation uses SLSQP for feasible-power proposals and L-BFGS-B for tangent points, followed by rational repair and independent bound reconstruction. The 44-case power-transfer experiment below is separate from this algorithm. Network pricing uses rectangular partitions; Theorem 3's algebraic oracle remains separate.

D. Performance measures

We report verified feasible values $L/\ln 2$, global gaps $U - L$ in nat, radiated power and target attainment. An increase in L represents an achieved-rate improvement; a decrease in U tightens the bound on remaining improvement. Matched comparisons use common channels, constraints and initialization. We distinguish the callable bound-construction times from the broader timing scopes of the earlier allocation methods.

TABLE I
CERTIFICATION BOUNDARY ON THE STORED RAY-TRACED CASES. THE $M = 8$ RESULT IS THE CERTIFIED BENCHMARK; THE $M = 2$ RESULT IS A CROSS-CHECK AND IS NOT USED FOR THE CDL RATE CLAIM.

Setting	Cases	Gap ≤ 0.01 nat	Remaining gap
$M = 8$	32	27	0.01–1.43 nat
$M = 2$	32	10	0.25–30 nat

E. Finite-box certification on ray-traced channels

We evaluate (18) on all 32 networks, using the same stored channels, constraints and grid anchors. The baseline is the unmodified configuration-dual repair of Proposition 1, supplied archived grid prices. It uses midpoint generic rectangular bounds, with whole-quadrant tangents where eligible, and one table with at most 1200 splits. The integral method permits up to three tables with revised price and tangent proposals. Each producer receives 10 s, followed by a separate 15-s checking allowance; imports, archive loading, fallback setup and I/O are excluded. Each method is run once per case.

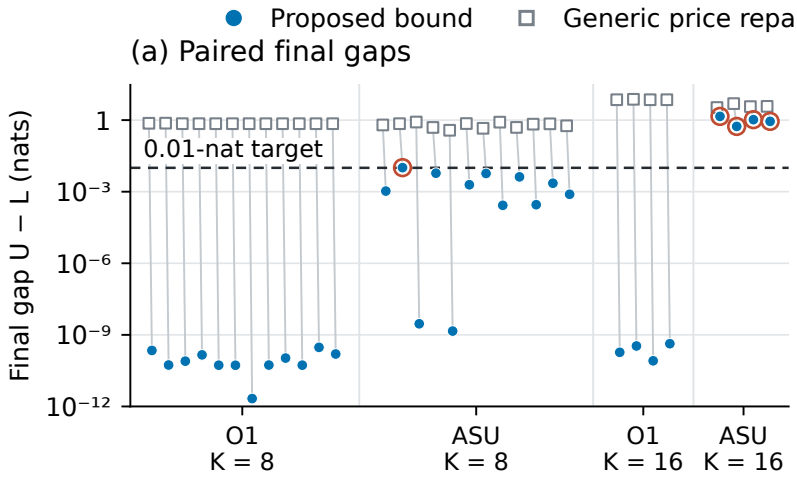
Fig. 2 shows final verified gaps $U - L \leq 0.01$ nat on 27/32 cases, versus 0/32 for generic repair. The median paired reduction in $U - L_{\mathrm{grid}}$ is 97.82%, using the same grid lower for both methods in each case. This measures upper-bound tightening separately from feasible-rate improvement. All sixteen O1 cases and eleven ASU cases meet the target; one eight-user and four sixteen-user ASU cases remain above it. Median construction/checking times are 1.429/0.597 s, versus 1.237/0.463 s. The tighter bounds thus come with higher overall median callable times.

At 0.1 W, median received SNRs are 47.24 dB in O1 and 16.42 dB in ASU over 896 user/resource links per scene, retaining 24 zero-gain ASU links. The quadrant-or-box test qualifies 5184/5184 O1 and 1663/5184 ASU pair/resource boxes, compared with 51/5184 and 471/5184 under the whole-quadrant condition alone. These pooled counts weight sixteen-user cases more heavily. The high-SNR O1 rectangles are certifiably concave despite large channel correlations.

The median achieved-rate gain over Grid is 0.04035%; only 3/32 cases exceed 1%. Against the earlier Full/pricing allocations, the median is 0.03555%, with only 2/32 above 1%. The new uppers place every earlier O1 allocation within 1.382×10^{-7} nat of the continuous integral optimum and bound possible improvement of the earlier Full allocations by 0.725% across all eight-user cases. On four matching physical inputs, twenty historical T/W/S/Fu-SCA/Thu-IDE records provide additional context: the saved SLSQP rates differ from the new rates by at most 3.92×10^{-8} nat. Their timing scopes differ, and we perform no rate-superiority or equivalence test. Supplementary Section S11 gives all outcomes, the two retained historical W verification timeouts, and the remaining budget-gap limitations.

VI. RECEIVE-DIMENSION REGIMES AND PAIRING HEADROOM

The original benchmark uses $M = 8$ receive dimensions, $K = 8$ or 16 users and two users per orthogonal resource.



(b) Target certificates

	Proposed bound	Generic price repair
O1, K=8	12/12	0/12
ASU, K=8	11/12	0/12
O1, K=16	4/4	0/4
ASU, K=16	0/4	0/4
All cases	27/32	0/32

Fixed-anchor $U - L_{\text{grid}}$
median reduction: 97.82%

Fig. 2. Finite-box/integral bounds versus archived configuration-dual repair on all 32 ray-traced cases. (a) Paired final gaps use each method's delivered feasible lower L; rings identify the five misses. (b) Target counts at 0.01 nat. The 97.82% statistic is the median paired reduction above the common grid lower L_{grid}, distinct from the final gaps in panel (a).

Before adding a new algorithm, we asked whether that benchmark can distinguish pairing methods at all. We computed, for every stored case, the best resource-labelled assignment at the stored grid powers and compared it with the grid assignment. The fixed-power assignment is solved exactly by the subset recurrence of Section VII. The result is unambiguous: in all 32 cases the grid assignment is already optimal at its own powers. The pairing gain is zero to numerical precision.

This is not a property of the channel data alone. It is a property of the receiver dimension. With two users per resource and $M = 8$, the MMSE receiver has enough spatial degrees of freedom to suppress the partner's signal almost completely. Changing the pairing then changes the rate very little. To isolate the mechanism, we truncated the same stored channel vectors to $M = 2$, $M = 4$ and $M = 8$ receive coordinates and repeated the exact assignment calculation at uniform power. No case was selected by its outcome.

Fig. 3 shows the result. At $M = 8$, the median pairing gain is zero, and only 3.1% of cases exceed 1%. At $M = 4$, the median gain is 2.61%, and 71.9% of cases exceed 1%. At $M = 2$, the median gain is 8.87%, and 93.8% of cases exceed 1%. The same channels therefore contain substantial pairing headroom; the $M = 8$ configuration hides it.

The certified headroom tells the same story. At $M = 8$, the O1 scene has a median received SNR of 47.2 dB, and every archived O1 allocation is within 1.38×10^{-7} nat of the continuous optimum. The median certified headroom over the grid is 5.4×10^{-10} in O1 and 0.75% in ASU. At $M = 2$, the fixed-power pairing gain is large enough to support a meaningful rate comparison. This is the regime we use for the exact search. The standardized 3GPP CDL benchmark of Section VIII is generated directly in this regime and is used for the rate claim; the ray-traced channels remain a cross-check.

The regime analysis also changes the interpretation of the earlier rate results. The $M = 8$ benchmark is useful for certification: the finite-box test qualifies every O1 rectangle, and the bound closes to 10^{-9} nat in the O1 cases. It is not useful for showing that a pairing algorithm improves the rate, because there is no pairing headroom to improve. A flat rate comparison at $M = 8$ is evidence of near-optimality, not evidence that the algorithms are equivalent in general.

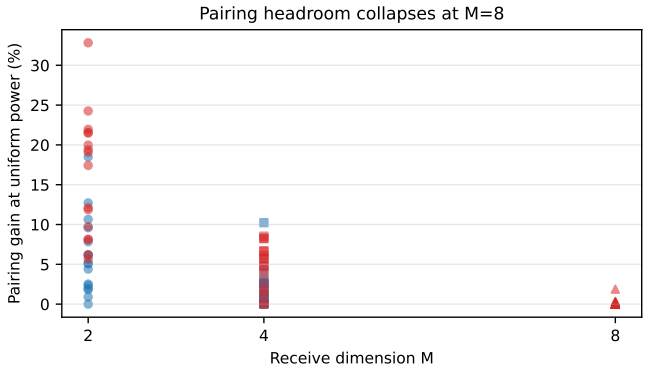


Fig. 3. Pairing gain at uniform power on the same 32 stored cases, after truncating the channel vectors to M receive coordinates. The gain is the exact best fixed-power assignment relative to the grid assignment. At $M = 8$ the median is zero; at $M = 4$ and $M = 2$ it is 2.61% and 8.87%. No case was selected by outcome.

VII. Exact Assignment and Convex Power Refinement

The certificate of Section III answers how far a schedule can be from the optimum. It does not by itself produce a better schedule. We therefore use the same pair-rate structure to solve the two subproblems exactly or to local optimality, and alternate them.

A. Exact fixed-power assignment

Fix the power vector p. For a pair $e = \{u, v\}$ on resource s, let

$$w_{uvs} = q_{r_{uvs}}(g_{us}p_u, g_{vs}p_v) \tag{1}$$

be the pair rate in natural-log units. The assignment problem is to choose one pair for every resource so that every user is served once. Let $V(A)$ be the best value for the user set A, with $|A|$ even. The recurrence is

$$\begin{aligned} V(\varnothing) &= 0, \\ V(A) &= \max_{\{u,v\}\subseteq A} \left[V(A \setminus \{u, v\}) + w_{uv,\,|A|/2-1}\right]. \end{aligned} \tag{2}$$

The resource index is $|A|/2 - 1$ because the recurrence fills resources in a fixed order. The recurrence is exact: every feasible schedule is a sequence of disjoint pairs, and every sequence is considered once. It uses $O(2^K)$ states and $O(K^2)$ transitions per state. For $K = 16$, the state space is 65 536 and the recurrence is negligible on a single CPU. For $K = 32$, the state space is too large; the implementation then falls back to matching and we report the boundary.

The recurrence differs from the integral bound of Section III in one respect. The bound uses price-derived upper bounds H_{uvs} for each pair and resource. Here the weights are the actual pair rates at the current powers. The same code therefore produces either a certified upper bound or an exact feasible assignment, depending on the weights.

B. Power refinement at a fixed assignment

With the assignment fixed, the remaining problem is

$$\begin{aligned} \max_{p} \quad & \sum_{s=1}^{S} q_{r_{e_s,s}}(g_{u_s,s}p_{u_s}, g_{v_s,s}p_{v_s}) \\ \text{s.t.} \quad & p_{\min} \le p_i \le p_{\max}, \qquad \sum_i p_i \le P. \end{aligned} \tag{3}$$

On a rectangle where the finite-box test of Section III holds, the pair rate is concave and the power problem is a convex program. Elsewhere we use sequential quadratic programming with the analytic gradient of the pair rate and exact repair of the power vector. The repair clips each power to the box and removes any excess budget from the largest slacks, so every returned vector is feasible.

The power step is local. To reduce the dependence on the starting point, we use one deterministic start and seven randomized starts. The deterministic start uses sequential pairs and uniform power P/K. Each randomized start uses a random permutation of the users and a feasible random power vector. For every start, we alternate the exact assignment and the power refinement until neither the assignment nor the power vector changes by more than 10^{-6}, or a wall-clock limit is reached. We retain the best feasible schedule. The complete procedure is summarized in Table II.

C. Relation to the certificate

The exact search is not a certificate. The certificate uses rectangular upper bounds and a subset recurrence with price weights; the exact search uses actual pair rates and a subset recurrence with feasible weights. The two share the recurrence and the pair-rate table, but their outputs answer different questions. The certificate returns an interval $[L, U]$; the exact search returns a feasible schedule and a rate. When the certificate is tight, the two agree; when it is loose, the exact search can still improve the rate, but the improvement is not globally certified. Section VIII reports both outcomes.

TABLE II
EXACT-SEARCH PROCEDURE.

1. Build the pair-rate table $w_{uvs} = q_{r_{uvs}}(g_{us}p_u, g_{vs}p_v)$ at the current powers.
2. Solve (2) exactly; obtain the resource-labelled assignment.
3. Fix the assignment and refine all powers by SQP with exact repair.
4. Repeat steps 1–3 until the assignment and powers stabilize.
5. Repeat from one uniform and seven randomized starts; retain the best feasible schedule.

TABLE III
CDL PARAMETERS AND GENERATED STATISTICS. NORMALIZED CLUSTER DELAYS ARE SCALED TO A FIXED 100 NS RMS DELAY SPREAD; THE ANGULAR SPREAD IS THE PROFILE AZIMUTH SPREAD c_{ASA}; XPR IS THE NOMINAL SINGLE-POLARIZATION VALUE.

Profile	Delay scale	c_{ASA}	XPR	r med.	SNR (dB)
CDL-A	100 ns	11°	10 dB	0.46–0.56	12.2–13.9
CDL-B	100 ns	22°	8 dB	0.49–0.52	12.2–13.9
CDL-C	100 ns	15°	7 dB	0.47–0.55	12.2–13.9

VIII. EXPERIMENTAL EVALUATION

A. Standardized CDL benchmark

We evaluate on the 3GPP TR 38.901 CDL-A, CDL-B and CDL-C models [30]. The cluster delays, powers and azimuth/elevation angles are taken from the standard tables. Each user is assigned a mean arrival angle; the cluster angles are offset by the profile's azimuth spread. We use a half-wavelength ULA with $M = 2$ receive coordinates, independent per-cluster phases and a 100 ns delay spread. The eight resources are mapped to the first eight tones of a 50 MHz, 32-tone grid. Every case is generated from a fixed seed; no case is selected by outcome.

The primary population has $K = 16$, $S = 8$ and 53 cases: three profiles, four angular separations and four seeds, plus five separation-sweep cases. The secondary population has $K = 8$, $S = 4$ and 24 cases. The power domain is $p_{\min} = 0.001$ W and $p_{\max} = 0.2$ W, with $P/K = 0.13$ W. The baseline grid is $\{0.05, 0.1, 0.2\}$ W, so it does not contain the continuous optimum. Every method starts from the same weak anchor: sequential pairing $(2s, 2s + 1)$ at uniform power P/K. The $K = 8$ runs receive 12 s and the $K = 16$ runs 10 s.

Table III summarizes the generated channels. The median squared correlation is about 0.5 and the 90th percentile is about 0.92; the median received SNR at 0.1 W is about 13–14 dB. The three profiles therefore cover a correlated, interference-limited regime rather than the high-SNR, nearly orthogonal regime of the earlier ray-traced benchmark.

TABLE IV
COMPARATOR DISCLOSURE.

Method	Status	Scope
Grid	full	finite grid; not a continuous bound
SLSQP	full	same box, budget and assignment family
I-WMMSE	adapted	common receiver and exact-once service
Matching–WMMSE	adapted	common receiver and exact-once service
Proposed	full	exact assignment + SQP power

TABLE V
CDL SUM RATE IN NAT, MEAN/MEDIAN.

Population	Grid	SLSQP	I-WMMSE	Matching	Proposed
$K = 16$, all	53.60/53.55	47.46/46.24	45.49/44.37	53.78/53.67	**54.56/54.39**
$K = 16$, A	54.16/53.80	46.16/45.18	44.24/43.19	54.45/54.06	**55.14/54.65**
$K = 16$, B	53.26/52.75	48.51/47.71	46.90/46.40	53.28/52.89	**54.21/53.64**
$K = 16$, C	53.20/52.71	48.09/47.48	45.74/44.73	53.39/52.81	**54.15/53.59**
$K = 8$, all	25.87/26.82	26.45/27.18	24.56/23.71	26.34/27.17	**26.52/27.41**

B. Methods and disclosure

The grid baseline solves the finite-grid integer program and its LP relaxation. SLSQP performs a full assignment search with continuous power refinement. I-WMMSE adapts the weighted-MMSE updates of [5] to the common receiver and exact-once service. Matching–WMMSE combines the matching step of [4] with the receiver and power updates of [5]. The proposed method is the exact assignment plus SQP power search of Section VII. Table IV separates full implementations from adaptations.

C. Metrics and statistics

The primary endpoint is the paired sum-rate difference between the proposed method and the best common-model baseline on the same case. We report the median and mean relative difference, the absolute difference in nat and in gross occupied-tone spectral efficiency, a whole-case bootstrap 95% interval with 10 000 resamples, and win/tie/loss counts. The tie band is 10^{-4} bit/s/Hz, above the numerical noise floor. The relative and absolute effects are reported separately because the sum rate grows with K.

D. Rate results

Table V reports the mean and median rates. At $K = 16$ the proposed method has the highest mean and median in every profile. At $K = 8$ it is essentially tied with SLSQP and matching–WMMSE, and clearly above the grid and I-WMMSE. Fig. 4 shows the distributions.

Table VI gives the paired comparison. At $K = 16$ the proposed method beats the best common-model baseline in all 53 cases. The median gain is 1.31% with a 95% interval $[1.19, 1.42]\%$; the mean is also 1.31%. The absolute median gain is 0.129 bit/s/Hz. At $K = 8$ the gain over SLSQP is small but positive: median 0.044%, interval $[0.106, 0.485]\%$ on the per-case ratio, with 13 wins, 10 ties and one loss. The absolute median gain is 0.0017 bit/s/Hz, which is negligible in practice. The K=8 result is therefore an efficiency result, not a rate result; the interval is on the mean relative difference, and the median lies below its lower bound because the distribution is right-skewed.

TABLE VI
PROPOSED MINUS EACH COMMON-MODEL BASELINE ON CDL. INTERVALS ARE WHOLE-CASE BOOTSTRAP 95% INTERVALS ON THE MEAN RELATIVE DIFFERENCE.

Population	Baseline	Median	Mean	95% CI	W/T/L	n
$K = 16$	best per case	1.31%	1.31%	$[1.19, 1.42]\%$	53/0/0	53
$K = 16$	matching	1.31%	1.48%	$[1.28, 1.69]\%$	53/0/0	53
$K = 16$	grid	1.88%	1.80%	$[1.76, 1.83]\%$	53/0/0	53
$K = 16$	SLSQP	13.70%	15.19%	$[13.63, 16.87]\%$	53/0/0	53
$K = 16$	I-WMMSE	19.49%	20.31%	$[18.40, 22.32]\%$	53/0/0	53
$K = 8$	SLSQP	0.044%	0.274%	$[0.106, 0.485]\%$	13/10/1	24
$K = 8$	matching	0.407%	0.731%	$[0.434, 1.044]\%$	22/2/0	24

TABLE VII
30-S BASELINE SENSITIVITY ON 12 CDL CASES. PROPOSED USES 10 S.

Comparison	Median	Mean	95% CI
Proposed − matching (30 s)	1.07%	1.15%	$[0.87, 1.45]\%$

Fig. 5 shows the per-case absolute gain over the best baseline. All 53 $K = 16$ cases are positive; the dashed line marks 0.05 bit/s/Hz. The gain is not driven by a few outliers: the distribution is concentrated between 0.05 and 0.27 bit/s/Hz. Fig. 6 breaks the result down by profile; all three profiles give positive intervals.

E. Attribution and runtime

Fig. 7 separates assignment and power on the CDL cases. Assignment alone gives a mean 54.8% gain over the uniform anchor, power alone gives 17.3%, and the joint method gives 54.9%. The gain is therefore predominantly from the exact assignment; the SQP power step adds a small refinement. This is consistent with the model: at $M = 2$ and $K = 16$, pairing is the main discrete lever, while the power step improves the selected pairs but does not change the ranking.

Fig. 8 shows the wall-clock times. The proposed method is the fastest arm at both sizes. At $K = 8$ its median time is about 0.057 s, versus about 12 s for SLSQP and 3.9 s for matching–WMMSE, a speedup of about 211 and 69 times by the ratio of medians. At $K = 16$ it is about 5.5 s, versus about 10 s for SLSQP and matching–WMMSE. The speedup is not obtained by a looser stopping rule; the rate comparison uses the same box and budget. A separate sensitivity gives matching–WMMSE and SLSQP 30 s on 12 CDL cases. Matching–WMMSE improves, but the proposed 10 s result remains ahead in all 12 cases, with a median gain of 1.07%. The baseline-strength concern is therefore not driving the result.

F. Ray-traced cross-check

The earlier DeepMIMO O1/ASU study is retained only as a ray-traced cross-check. At $M = 8$ it has zero fixed-power pairing gain in all 32 cases, which explains why the earlier

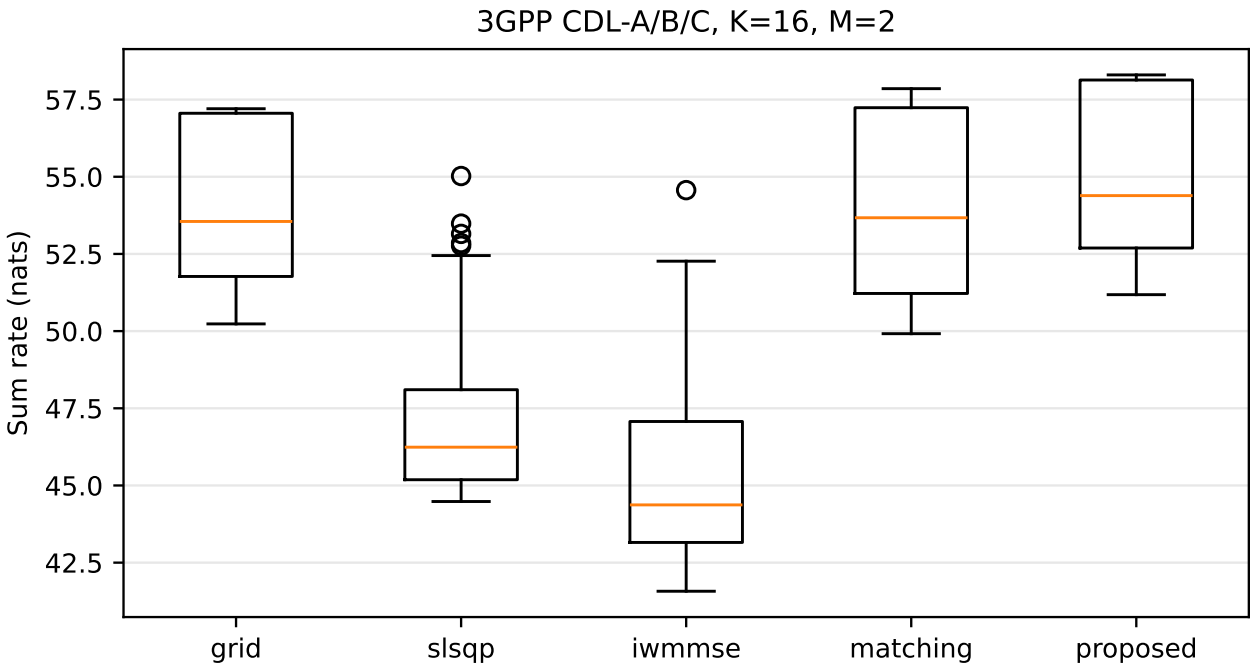


Fig. 4. CDL sum-rate distributions at $K = 16$, $M = 2$. The proposed exact search has the highest median and the tightest upper tail.

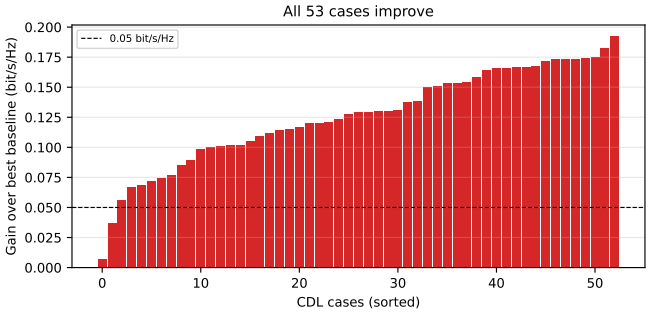


Fig. 5. Absolute gain over the best common-model baseline on the 53 CDL $K = 16$ cases, sorted. Every case is positive; the dashed line marks 0.05 bit/s/Hz.

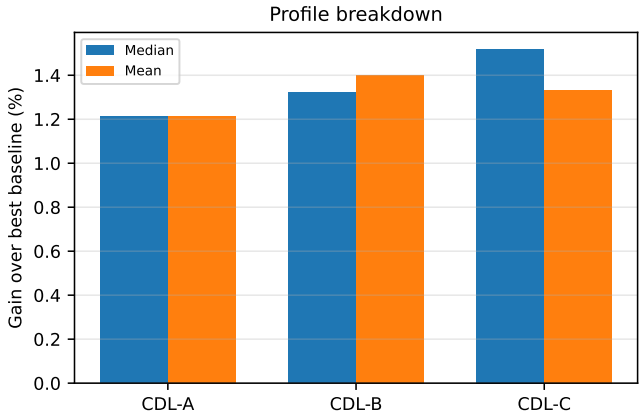


Fig. 6. Median and mean gain by CDL profile. All three profiles are positive, with the largest median on CDL-C.

rate comparison was flat. At $M = 2$ the same channels show pairing headroom, but the $K = 16$ rate result there rests on eight cases and is not used for the headline. The certified $M = 8$ result remains 27 of 32 cases at 0.01 nat with a 97.82% median fixed-anchor gap reduction. The CDL benchmark replaces DeepMIMO for the rate claim because it is standardized, seed-reproducible and has 53 cases.

IX. Limitations and Discussion

The results separate three claims. At $M = 8$, the finite-box/integral certificate is strong: 27 of 32 stored ray-traced cases close to 0.01 nat, with a 97.82% median reduction in the fixed-anchor gap. That result concerns certification, not rate superiority. At $M = 2$ and $K = 16$, the exact search improves the strongest common-model baseline by a median 1.31%, or 0.129 bit/s/Hz, on 53 standardized 3GPP CDL cases, with 53 wins in 53 cases. That result concerns rate, not certification. The paper does not claim certified rate superiority.

The CDL benchmark is standardized and seed-reproducible, but it is still a model, not a measurement. The cluster tables are taken from TR 38.901 and the generator is included with the artifact; nevertheless, a hardware campaign would be needed before treating the 1.31% median as a deployment guarantee. The CDL rate result is also not globally certified. We did not run the finite-box certificate on the 53 CDL cases, and the earlier $M = 2$ ray-traced certificate was loose on 22 of 32 cases. The rate gain is feasible and verified, but its distance from the CDL optimum is not enclosed by a tight interval.

At $K = 8$, the proposed method matches SLSQP and

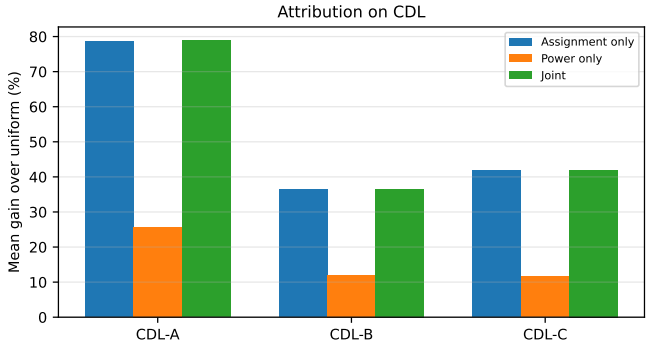


Fig. 7. Mean gain over the uniform anchor on the CDL cases. Assignment-only uses the exact assignment at uniform power; power-only refines powers on the sequential assignment; joint is the proposed method. The bars are separate arms, not additive components.

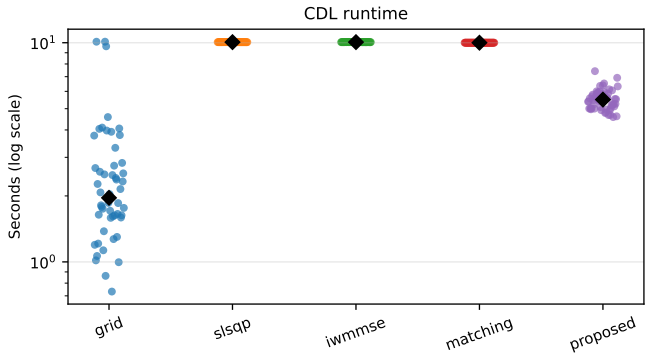


Fig. 8. CDL wall-clock time on a log scale. The proposed exact search is the fastest arm at both $K = 8$ and $K = 16$.

slightly exceeds it in the median, but the absolute gain is about 0.0017 bit/s/Hz. That is not a practical rate advantage; the meaningful result at $K = 8$ is the roughly 211-fold runtime reduction. The K=16 result is the rate result.

The comparison uses common-model baselines. I-WMMSE and matching–WMMSE are adaptations of their source algorithms to the common receiver and exact-once service, not full reproductions of the source systems. SLSQP and I-WMMSE fall below the grid at K=16 because their assignment searches reach the time limit; the exact DP solves the assignment for the full user set. We report this explicitly and treat matching–WMMSE as the strongest common-model baseline at K=16.

The model is nominal. Channels are static, perfectly known and ideally synchronized. The formulation uses a common receiver, exact-once service, two users per resource and a sum-rate objective. It does not include channel estimation error, aging, fronthaul limits, SIC, per-user QoS or fairness constraints. A sum-rate gain can reduce the minimum-user rate; we do not claim service fairness.

The CDL generator uses the standard cluster tables with independent per-cluster phases and a simplified ULA response. It does not reproduce every ray-level detail of a full 3GPP simulator. The validation table reports the generated correlation and SNR ranges, and the code is included, but a full Sionna or QuaDRiGa cross-check remains future work. The ray-traced DeepMIMO result is retained only as a cross-check: it explains the $M = 8$ null result and supports the receive-dimension mechanism, but it is not used for the CDL rate claim.

X. Conclusion

We studied full-service MMSE pair scheduling through fixed-dimensional hardness, sharp pairwise curvature and finite rectangular pricing. The finite-box/integral certificate closes to 0.01 nat on 27 of 32 stored ray-traced networks at $M = 8$, with a 97.82% median reduction in the fixed-anchor gap.

A receive-dimension audit explains the earlier flat rate comparison. At $M = 8$, MMSE reception removes the two-user interference, and the fixed-power pairing gain is zero in all 32 cases. Headroom appears at smaller receive dimensions: the median pairing gain is 2.61% at $M = 4$ and 8.87% at $M = 2$. On 53 standardized 3GPP TR 38.901 CDL-A/B/C cases at $K = 16$ and $M = 2$, the same exact search improves the strongest common-model baseline by a median 1.31%, or 0.129 bit/s/Hz, with 53 wins in 53 cases and a 95% interval $[1.19, 1.42]\%$. At $K = 8$ it matches SLSQP with a negligible absolute gain while running about 211 times faster. These CDL rate gains are feasible and verified but not globally certified; the certified result remains the $M = 8$ bound. The boundary between certified near-optimality and regime-dependent rate gains is the main message of the paper.

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Data Availability

The accompanying review artifact contains the experimental records, analysis scripts, figure sources, the 3GPP CDL generator and the independent verification code. Supplementary Section S10 describes reproduction and the archived studies; Section S13 documents the CDL benchmark. Raw DeepMIMO scenarios are obtained separately. The public repository will be released upon acceptance.

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